

# Francesco Tornabene

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## SUMMARY

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I'm currently a PhD student at Gran Sasso Science Institute, L'Aquila. My interests concern mathematical general relativity, Riemannian geometry, and optimal transport. In particular, my PhD project involves boundedness and decay of linear waves on Lorentzian manifolds.

## WORK EXPERIENCE

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### Scholarship at Università di Pavia

Oct 2023 - Jun 2024

I worked on existence, uniqueness, and characterization in 1D of Wasserstein barycenters with possibly negative weights, with professor M. Veneroni; moreover, I worked on the use of machine learning for spectroscopy with prof. S. Gualandi.

### Tutor at Università di Pavia and Università di Bergamo

2021-2024

I worked as a tutor at both Università di Pavia and Università di Bergamo for the courses "Analysis 1" and "General Mathematics". My work involved preparing and solving exercises, as well as individual tutoring.

### Teacher at High School "Liceo Scientifico Niccolò Copernico (Pavia)"

Apr-May 2022

I was a full time teacher of mathematics and physics for two months. In particular, I gave frontal lectures and prepared calibrated written exams on elementary topics.

## EDUCATION

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2024 - present    PhD (Mathematics) at **GSSI**

2020 - 2022    Master's Degree in Mathematics at **Università di Pavia** (110L/110);  
Thesis: *Metrics on Triangulated Categories and Applications to Derived Categories of Schemes*  
Advisor : Prof. A. Canonaco

2017 - 2020    Bachelor's Degree in Mathematics at **Università di Pavia** (110L/110)  
Thesis: *The Fundamental Theorem of Calculus in the Theory of Integration of Henstock-Kurzweil*  
Advisor: Prof. G. Schimperna

## PUBLICATIONS

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F. Tornabene, M. Veneroni & G. Savaré (2025). "Generalized Wasserstein barycenters". In: *Applied Mathematics & Optimization* 92.68. DOI: <https://doi.org/10.1007/s00245-025-10351-6>.

## SKILLS

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Good proficiency in Python, especially in the context of machine learning.

Teaching experience, both as frontal lecturer and individual tutor.

I especially enjoy inventing interesting and well-suited mathematical problems for students of all levels.